Name: Shshank Pratap Roy **Section:** A **Subject:** OOPS With Java **Rollno.:** 2503201450138

Assignment 1

Ques1: Write a Java Program that takes input as an array of integers and returns the index corresponding to the lowest.

LowestIndex.java

```java
import java.util.Scanner;

public class LowestIndex {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        // taking size of array
        System.out.print("Enter number of elements: ");
        int n = sc.nextInt();

        int[] arr = new int[n];

        // taking array input
        System.out.println("Enter the elements:");

        for(int i = 0; i < n; i++){
            arr[i] = sc.nextInt();
        }

        // letting first element as the smallest
        int min = arr[0];
        int index = 0;

        // find lowest element index
        for(int i = 1; i < n; i++){
            if(arr[i] < min){
                min = arr[i];
                index = i;
            }
        }
        System.out.println("Index of the lowest element is: " + index);
    }
}
```

Output:

```
$ javac LowestIndex.java

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$ java LowestIndex
Enter number of elements: 4
Enter the elements:
12
33
5
11
Index of the lowest element is: 2
```

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Ques2: Write a Program to calculate the hourly salary of 3 employees. Write a method that takes the base pay and hours worked as parameters, and prints the total pay or an error. Write a main method that calls this method for each of these employees. If the number of hours is greater than 60, print an error message.

EmployeeSalary.java

```java
public class EmployeeSalary {

    public void calculatePay(double basePay, int hoursWorked) {

        if (hoursWorked > 60) {
            System.out.println("Error: Employee worked more than 60 hours");
        }
        else {
            double totalPay = basePay * hoursWorked;
            System.out.println("Total Pay = " + totalPay);
        }

    }

    public static void main(String[] args) {

        EmployeeSalary obj = new EmployeeSalary();

        obj.calculatePay(10.5, 40);
        obj.calculatePay(12.0, 50);
        obj.calculatePay(9.5, 65);

    }
}
```

Output:

```
$ javac EmployeeSalary.java

User@DESKTOP-6D1E00S MINGW64 ~/Desktop/java/Assignment
$ java EmployeeSalary
Total Pay = 420.0
Total Pay = 600.0
Error: Employee worked more than 60 hours
```

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Ques3: Write a Java Program to find the Armstrong number.

ArmstrongNumber.java

```java
import java.util.Scanner;

public class ArmstrongNumber {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int num = sc.nextInt();

        int original = num;
        int remainder;
        int sum = 0;

        while (num != 0) {

            remainder = num % 10;
            sum = sum + (remainder * remainder * remainder);
            num = num / 10;

        }

        if (sum == original) {
            System.out.println(original + " is an Armstrong number");
        }
        else {
            System.out.println(original + " is not an Armstrong number");
        }

    }
}
```

Output:

```
$ javac ArmstrongNumber.java

User@DESKTOP-6D1E00S MINGW64 ~/Desktop/java/Assignment
$ java ArmstrongNumber
Enter a number: 153
153 is an Armstrong number
```

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Ques4: Write a Java Program to print the Floyds Triangle.

FloydTriangle.java

```java
import java.util.Scanner;

public class FloydTriangle {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of rows: ");
        int n = sc.nextInt();

        int num = 1;

        for (int i = 1; i <= n; i++) {

            for (int j = 1; j <= i; j++) {

                System.out.print(num + " ");
                num++;

            }
            System.out.println();
        }
    }
}
```

Output:

```
$ javac FloydTriangle.java

User@DESKTOP-6D1E00S MINGW64 ~/Desktop/java/Assignment
$ java FloydTriangle
Enter number of rows: 5
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

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Ques5: Program to remove duplicate element in an array.

RemoveDuplicate.java

```java
import java.util.Scanner;

public class RemoveDuplicate {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of elements: ");
        int n = sc.nextInt();
        int[] arr = new int[n];
        System.out.println("Enter array elements:");

        for(int i = 0; i < n; i++){
            arr[i] = sc.nextInt();
        }
        System.out.println("Array after removing duplicates:");

        for(int i = 0; i < n; i++){

            boolean isDuplicate = false;

            for(int j = 0; j < i; j++){
                if(arr[i] == arr[j]){
                    isDuplicate = true;
                    break;
                }
            }
            if(isDuplicate==false){
                System.out.print(arr[i] + " ");
            }
        }
    }
}
```

Output:

```
$ javac RemoveDuplicate.java

User@DESKTOP-6D1E00S MINGW64 ~/Desktop/java/Assignment
$ java RemoveDuplicate
Enter number of elements: 5
Enter array elements:
1
5
5
5
3
Array after removing duplicates:
1 5 3
```